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FiFrauD: Unsupervised Financial Fraud Detection in Dynamic Graph Streams

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Given a stream of financial transactions between traders in an e-market, how can we accurately detect fraudulent traders and suspicious behaviors in real time? Despite the efforts made in detecting these fraudsters, this field still faces serious challenges, including the ineffectiveness of existing methods for the complex and streaming environment of e-markets. As a result, it is still difficult to quickly and accurately detect suspected traders and behavior patterns in real-time transactions, and it is still considered an open problem. To solve this problem and alleviate the existing challenges, in this article, we propose FiFrauD, which is an unsupervised, scalable approach that depicts the behavior of manipulators in a transaction stream. In this approach, real-time transactions between traders are converted into a stream of graphs and, instead of using supervised and semi-supervised learning methods, fraudulent traders are detected precisely by exploiting density signals in graphs. Specifically, we reveal the traits of fraudulent traders in the market and propose a novel metric from this perspective, i.e., graph topology, time, and behavior. Then, we search for suspicious blocks by greedily optimizing the proposed metric. Theoretical analysis demonstrates upper bounds for FiFrauD's effectiveness in catching suspicious trades. Extensive experiments on five real-world datasets with both actual and synthetic labels demonstrate that FiFrauD achieves significant accuracy improvements compared with state-of-the-art fraud detection methods. Also, it can find various suspicious behavior patterns in a linear runtime and provide interpretable results. Furthermore, FiFrauD is resistant to the camouflage tactics used by fraudulent traders.

CCS Concepts: • **Security and privacy** → **Systems security**; • **Applied computing** → **Secure online transactions**; • **Information systems** → **Decision support systems**;

Additional Key Words and Phrases: Dynamic graph, fraud detection, market manipulation detection, stream learning, unsupervised learning

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1 INTRODUCTION

How can we effectively detect online traders who manipulate e-markets by buying or selling to achieve their desired profit by deceiving other traders? More challenging, how can we detect fraudulent traders or market manipulators in a stream of online financial transactions in real time? The problem of manipulation, the intensity of which is increasing day by day, has become a

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significant challenge in financial e-markets because it provides big profits for the manipulators to the extent that during the manipulation in 2010, for example, there was a loss of about \$1 trillion in the NASDAQ market [1]. In another example, a person obtained illegal profits of at least \$20.9 million through different accounts [2]. A report states that 20,000 investors lost over \$30 million due to a manipulation scheme in which a group of traders conspired to artificially increase a company's share price [3].

The monetary losses caused by manipulation activities in financial markets are just the tip of the iceberg, as they also erode participants' trust and lead them to trade in other markets [4]. Thus, these activities may result in a liquidity reduction, higher trading expenses, and, eventually, a decline in job creation and economic growth [5, 6]. Therefore, effectively detecting manipulation activities is necessary and vital to prevent their harmful consequences [7]. For this reason, various studies have focused on the manipulation detection problem and have achieved acceptable results [8–11]. Still, this field faces serious new challenges requiring broader and deeper studies. Some of these challenges are as follows.

- **Existence of Complex Relationships between Traders:** In financial markets, the relationships among traders are complex because fraudsters engage in camouflage by trading with both honest and fraudulent traders in an attempt to appear authentic. Honest traders, on the other hand, usually trade with other honest traders and occasionally with fraudsters. This phenomenon is also known as “heterophily,” which means trading with both fraudulent and honest groups.
- **Existence of Correlation and Dependence between Trading Behaviors:** Traders influence each other's behavior, which automatically leads to immediate or emotional buys/sales.
- **Streaming Nature of Trading Data:** E-markets generate an ever-increasing volume of records that come in quickly. Therefore, analyzing these data to detect manipulations is demanding and challenging.
- **Lack of Labeled Data:** Only a tiny percentage of trading behaviors are considered manipulative, and few of these manipulative cases are pursued by regulatory authorities.

Despite the different challenges listed above, the existing manipulation detection approaches [4, 10, 12–15] are designed for static records, perform the detections based on classic machine learning methods, and ignore the complex relationships and high dependence between traders. Most of them are supervised or semi-supervised, which require enough training datasets with labeled data records, and do little analysis of the behavior of market manipulators, focusing on the detection model. Therefore, these approaches are not responsive to market needs, fail to deal with streaming data, and will not be able to detect manipulations.

On the other hand, the methods that detect anomalies based on graphs [16–19] or outlier detection [20–22] do not account for the traits of manipulative behavior, leading to lower accuracy in detection or limited applicability in this domain. Therefore, despite the efforts made in this area, it is still difficult to quickly and accurately identify suspicious traders and behavior patterns in real-time transactions. Thus, it is still an open problem. It seems that the problem of detecting **market manipulation (MM)** suffers from a severe gap in detection methods and is still in its infancy. This gap has been emphasized in review studies [7, 23–27] as well.

To solve this problem and alleviate the challenges mentioned above, we regard detecting traders engaged in manipulation as a problem of finding dense subgraphs, an unsupervised graph-based approach, in a stream of dynamic graphs. For this purpose, we analyze the main traits of manipulation behavior and define density based on a class of metrics. Finding densely connected groups of nodes is a primary graph-mining technique [28]. This technique optimizes a formally defined

metric over the entire dataset, taking advantage of the subgraph's internal structure [29, 30]. Subgraph mining has been widely studied due to its robustness in fraud detection [29, 31, 32]. However, it has not been employed for MM.

Therefore, in this article, we propose **FiFrauD (Financial-Fraud-Detection)**, a scalable unsupervised approach that depicts the behavior of manipulators in a transaction stream. In this approach, we specifically consider transaction relations between traders, turn these real-time transactions into a stream of graphs, and capture fraudulent traders by precisely utilizing density signals in graphs instead of using a supervised or semi-supervised learning method. Then, we propose a class of metrics to determine transaction suspiciousness from three perspectives: graph topology (structure), time, and behavior.

Contributions. The main contributions of our article are:

- **Streaming algorithm:** To our knowledge, this article is the first work that focuses on detecting fraudulent traders in a transaction stream. Our novel approach, FiFrauD, can quickly answer detection queries anytime based on the proposed metric.
- **Graph:** To our knowledge, FiFrauD is the first algorithm employing dense subgraph mining to detect fraudulent traders in financial markets.
- **Novel suspiciousness metric:** We define a class of suspiciousness metrics of traders using topology (structure), time (temporal bursts and drops), and behavior. This leads to the interpretability of the results.
- **Theoretical guarantee:** We show that fraudsters will be tracked if they do the market manipulation in less time than a theoretical bound considered for time or with the number of transactions less than the theoretical bound for the number of manipulative transactions. That means that, regardless of how the fraudsters manipulate the time or the number of their transactions, if the duration of the manipulation lasts less than a particular time or the number of their transactions is less than a specific limit, the fraudulent traders will be caught. Therefore, FiFrauD can obstruct fraudsters and increase their time cost and effort needed to manipulate a market.
- **Scalability:** The runtime of FiFrauD increases in a linear relationship with the number of edges of the transaction streams.

Outline. The rest of the article is organized as follows. Section 2 summarizes the related work. Financial fraud and its types in financial markets are defined in Section 3. Section 4 presents the key traits of MM and provides an informal definition of the problem. Section 5 describes the suspiciousness metric and an algorithm is proposed to detect fraud. Experiments are designed and conducted in Section 6. Section 7 discusses the results and implications. Our conclusions are presented in Section 7.

2 RELATED WORK

As this study aims to identify anomalous trading behaviors based on finding dense subgraphs in a stream of dynamic graphs, in this section, we review the approaches related to market manipulation detection and graph-based anomaly detection.

2.1 Market Manipulation Detection

Most of the studies detect the manipulation in a *supervised* manner. The dataset in these studies is often collected by examining manipulation cases reported by relevant organizations (such as the **Securities and Exchange Commission (SEC)**) and labeling the assets as manipulated or normal. Then, they classify the manipulated and normal assets by constructing a detection model based on this dataset [33]. For example, Diaz et al. [2] use decision trees (i.e., CART, C5.0, and QUEST) to construct a detection model and utilize indicators of returns, abnormal returns, liquidity, and

volatility in these models. According to their results, decision trees perform well in classifying manipulated and normal stocks, among which C5.0 has higher accuracy. Then, Golmohammadi et al. [34] work on the dataset of [2]. Based on that, they compare the performance of different supervised algorithms for classifying the stocks. According to the results, the **Naïve Bayes (NB)** algorithm performs best compared with **Classification and Regression Tree (CART)**, **Random Forest (RF)**, **Artificial Neural Network (ANN)**, **Support Vector Machine (SVM)**, and **K-Nearest Neighbor (k-NN)** techniques. By studying the manipulation cases prosecuted by CSRC, Wang et al. [9] generate a dataset of manipulation cases at the intraday level. They make **recurrent neural network (RNN)**-based ensemble models based on trading and characteristic features. The results show that using a combination of random subsets of features is better than using a single classifier in the classification of manipulated and non-manipulated stocks, and considering characteristic features will lead to improved results. A case study is conducted by Uslu and Akal [15] based on manipulated stocks of BIST to capture manipulation activities. The authors use different modeling algorithms, such as **Logistic Regression (LR)**, **Decision Tree (DT)**, k-NN, RF, SVM, and NB. The finding shows that their method has 91% F_1 -measure, 95% sensitivity, and 93% accuracy for detecting manipulations.

Some research uses a semi-supervised approach for manipulation detection; the detection model is constructed based on normal and non-manipulated data [35]. These studies synthetically generate manipulation cases and inject them into the normal data. Finally, the detection model is evaluated based on the dataset, including synthetic manipulations and normal cases. For example, Zhai et al. [11] try to detect manipulations in the stock market by identifying abnormal patterns with **Hidden Markov Model with Anomaly States (HMMAS)** and **One-Class SVM (OCSVM)** methods. They extract the main features related to the price using wavelength analysis and construct the model with these features along with the tick data from NASDAQ Market and London securities. Their model is evaluated with datasets including real normal data and artificial manipulations; the results show the appropriate performance of the model. In their study, Leangarun et al. [36] try to detect MM by integrating deep learning and **Generative Adversarial Network (GAN)** techniques. They analyze the **Limit Order Book (LOB)** reconstructed from the Thai Stock Exchange and develop a detection model by integrating **Long Short-Term Memory (LSTM)** and GAN. The model is evaluated by simulating the manipulation cases and using the accuracy metric, whose accuracy in supervising new data equals 68.1%.

Although these studies use supervised and semi-supervised approaches to detect MM, their approaches depend on labeled data and are not suitable for streaming data. This is because the data streams are numerous and evolving quickly; thus, it is impossible to intervene and label the data. As a result, operating in an unsupervised manner is frequently required. This study focuses on spotting manipulation activities in the streaming environment and unsupervised manner.

Another group also deals with **market manipulation detection (MMD)** using an unsupervised approach. For example, Rizvi et al. [37] identify price manipulation based on a combination of the **Dendritic Cell Algorithm (DCA)** and Kernel Density Estimate Algorithms. They try to detect two types of manipulations, pump-and-dump and spoofing, based on anomaly detection in stock prices using NASDAQ stock tick price data and artificial manipulations. The evaluation results regarding false alarms (0%–68%) and **Area under the Curve (AUC)** (84%–99%) indicate the proposed method's appropriate performance in the offline environment. Leangarun et al. [13] compare the effectiveness of autoencoders with GANs for identifying MM and offer a “MinMan-iMax” optimization technique. They show that five out of six actual manipulations are correctly identified by both models. Golmohammadi and Zaiane [4] propose **Contextual Anomaly Detection (CAD)** for detecting contextual anomalies in time series. The authors use the normal dataset of S&P stock prices at two levels of daily and weekly granularity and synthetically generate the

anomalies. Although their proposed method performs better than other baselines, such as Random Walk and k-NN, this method still suffers from low precision, i.e., high **False Positives (FPs)**. Chulamonthon and Tangamchit [12] employ a mix of supervised and unsupervised techniques. They analyze the **Stock Exchange of Thailand (SET)** data and use a combination of LSTM-based **AutoEncoder (LSTM-AE)** and LSTM to detect manipulation in the static environment. The authors contrast their model's performance with separately used LSTM and LSTM-AE for identifying six real manipulation cases and some synthetic manipulation cases. They conclude that combining deep learning methods leads to improved performance.

Some studies also use the approach of trading network analysis to recognize their trading behavior. These studies primarily seek to distinguish between manipulated and normal assets based on network structure. In these networks, the traders are represented by the nodes, and their transactions are represented by the edges. Shi et al. [38] construct a weighted undirected trading network for each stock on the China stock exchange and examine the clique structure (a complete subgraph) in these networks. They found that the number of cliques and their average and maximum weight in the manipulated stocks are higher than normal stocks. Sun et al. [14] build a multipart trading network based on day and cross-day trading activities for each stock in the China market. They found that, in terms of the relationship between traders' trading frequency and activity, the trading network of normal stocks behaves similarly to its random version, but this does not happen for manipulated stocks. Zhai et al. [10] construct a weighted directed trading network based on NYSE and CME tick data. Based on the price, volume, and structure of transactions in the network, they conclude that there is a closed cycle between colluding traders in the way that the collusions include orders with the features of short lifetimes, coordinated volumes, and viable prices. Based on these results, they detected collusion through a knapsack problem. This article has studied many related works, summarized in Tables 1 and 2 in Section A.3 of the appendix.

To overcome the constraints of supervised and semi-supervised methods, these studies were based on the unsupervised approach and achieved good results. However, it seems that they did not study the manipulation detection problem completely and deeply. Thus, some critical challenges in this field have not been solved yet. For example, in these studies, it is assumed that from the beginning, all the data are available and, therefore, they use batch learning methods to detect manipulation. The focus is mainly on developing detection methods, and little analysis has been done on the behavior of market manipulators to the extent that they detect the manipulations based on just one feature of the orders or some market statistical features. The relational nature of trading activities has not been taken into consideration because traders' trading behaviors on financial markets are not independent of one another but rather impact one another. Moreover, there are complicated relationships between them. Therefore, it is necessary to consider the dependencies and relationships in the data. In this regard, graph mining or trading network analysis methods assist in analyzing the relationships between traders and then modeling them. Nevertheless, few studies employ this approach to detect manipulations, and they also assume the trading networks to be static. The graph-based unsupervised method (i.e., dense subgraph detection) has attracted the attention of many researchers due to its high robustness in fraud detection. Therefore, this study aims to detect abnormal buying and selling behavior and traders based on the trading network analysis approach by finding dense subgraphs in a stream of dynamic trading graphs. For this purpose, the main traits of manipulation behaviors are analyzed and the density is defined based on a class of metrics. Table 1 shows distinctive dimensions between this study and previous studies in the field of MMD.

2.2 Graph-Based Anomaly Detection

Graphs provide a powerful mechanism for considering the related dependencies between data objects, and many graph-based methods have been developed for anomaly detection. *Graph-based*

Table 1. Distinctive Dimensions of This Study in Comparison with Existing Studies on Market Manipulation Detection

Reference	Scope		Model		Data		Approach			Graph		Label		Runtime	
	Ad-hoc	General-purpose	Black-Box	Behavior Analysis	Historical	Stream	Supervised	Unsupervised	Semi-supervised	Static	Dynamic	Synthetic	Real	Linear	Non-linear
Palshikar and Apte [39]	✓		✓		✓			✓		✓			✓		✓
Diaz et al. [2]	✓		✓		✓		✓						✓		✓
Golmohammadi et al. [34]	✓		✓		✓		✓						✓		✓
Golmohammadi and Zaiane [4]	✓			✓	✓			✓				✓		✓	
Zhai et al. [11]	✓			✓	✓				✓			✓			✓
Zhai et al. [10]	✓		✓		✓			✓		✓			✓		✓
Leangarun et al. [36]	✓		✓		✓				✓			✓			✓
Rizvi et al. [37]	✓		✓		✓			✓				✓			✓
Martínez Miranda et al. [40]	✓		✓		✓		✓						✓		✓
Chen et al. [41]	✓		✓		✓			✓					✓		✓
Wang et al. [9]	✓		✓		✓		✓						✓		✓
Shi et al. [38]	✓		✓		✓			✓		✓			✓		✓
Leangarun et al. [13]	✓		✓		✓			✓				✓	✓		✓
Uslu and Akal [15]	✓		✓		✓		✓						✓		✓
Chullamonthon and Tangamchit [12]	✓		✓		✓		✓	✓				✓	✓		✓
This article		✓		✓		✓		✓		✓	✓	✓		✓	

techniques can identify groups of fraudsters by discerning structural anomalies via density signals. Since frauds inevitably result in related connections or graph edges, it is difficult to avoid such identification. Recent research has demonstrated its effectiveness in a variety of anomaly detection tasks. *OddBall*, which finds novel patterns in the distribution of eigenvalues for anomaly detection, is used to identify anomalies in graphs [42]. By concentrating on bicliques, Braun et al. [43] construct a bipartite shop-card graph and specify the anomalousness score. To detect transactional fraud, Herrera-Semenets et al. [44] consider each transaction as an instance of a star graph. The authors use vector representations created from frequent subgraphs to train the classifiers. By integrating the traits that fraudsters often have, *SliceNDice*, as an axiomatic-based algorithm, detects them in Snapchat's advertising application [45].

Most existing works, due to the limitations and challenges stated in the previous section, study the fraud detection problem with an unsupervised method based on the *density of blocks* in adjacency matrices or multi-mode tensors. To improve their computations, various techniques such as one-step computation [46], reachable distance functions [47], hyper-class representation [48], fair representation [49, 50], high-dimensional data processing [51], and efficient computations [52, 53] were proposed. *CoreScope* detects the abnormal dense blocks in large graphs based on the k-core analysis [54]. Tensor decomposition is another basis for detecting dense blocks. *CrossSpot* estimates the block suspiciousness by using the Poisson model [28]. By maximizing the number of edges in blocks limited inside time periods, *CopyCatch* also detects lockstep activity [55].

To improve detection accuracy, existing research has focused on the suspiciousness of nodes or edges in real-world graphs. *Fraudar* identifies a group of suspicious nodes in dense subgraphs that

attempt to camouflage themselves by making random relationships with regular users. *Fraudar* mainly relies on structural features and defines a density function based on them. It calculates the suspiciousness of edges using the inverse logarithm of objects in degrees so that less weight is given to popular objects [56]. *HoloScope* is a framework suitable for bipartite graphs that spots suspicious reviewers based on temporal spikes, hyperbolic topology, and rating deviation. In order to identify dense subgraphs based on node anomaly scores, it uses singular vectors to find communities with strong connections as seeds [17]. *FlowScope* finds dense multi-partite structures on trading networks to detect money laundering [30]. By combining time, capital, cyclicity, and topology signals in a bipartite graph, *ANTICO* provides a metric to identify cash-out fraud behaviors in commercial banks' credit card business [31]. *AntiBenford* detects anomalies in cryptocurrency transaction networks based on Benford's law. It identifies a group of traders who make many transactions together and deviate greatly from Benford's law. According to the findings, *AntiBenford*'s subgraphs exhibit traits seen in illegal transactions, indicating that fraudsters coordinate abnormally in generating particular subgraph structures and carefully separating the money [29]. *AnoEdge* is a framework for detecting anomalous behaviors, such as network intrusions in which a large number of edges are created between source and destination nodes, in streaming graphs. It transforms the problem of dense subgraph detection into dense submatrix detection. To do this, it proposes a higher-order sketch data structure by extending the two-dimensional sketch to preserve the dense subgraph structure during the transformation. Then, it detects anomalies in this data structure [57]. *SpokEn* detects anomalies in large graphs by using correlations between eigenvectors to discover near-cliques and near-bipartite cliques [58]. It is worth mentioning that all of the above approaches study the *bipartite* graph.

Our work, which is about customizing the graph (that is, a unipartite one) for spotting fraudulent traders in e-markets, is different from the stated approaches. Table 2 compares FiFrauD and other related approaches (anomaly detection based on dense subgraph discovery). In the table, *camouflage* means that the detection methods are not sensitive to resistance or camouflage. As we can see in the table, FiFrauD is the only one that considers the full property list in a unified suspiciousness framework, including temporal spikes (sudden bursts and drops, and multiple spikes), topology, and coherent behavior. Also, FiFrauD is the first work that, for the purpose of MMD, identifies anomalies in dynamic graph streams and the first algorithm that employs dense subgraph mining for detecting fraudulent traders in financial markets.

3 FRAUD IN FINANCIAL MARKETS

Fraud and fraudulent behavior are defined in many different ways. The Association of Certified Fraud Examiners defines *fraud* as misusing corporate resources or assets for personal advantage [59]. According to the Oxford English Dictionary, *fraud* is an improper and criminal deception to obtain money or other benefits [60]. The FBI has one of the best classification schemes for financial crimes, depicted in Table 3, and classifies financial fraud into two levels [61].

Fraud in financial markets is one of the forms of financial fraud listed in Table 3. In the financial markets, "fraud" is referred to as "market manipulation," including several forms of market abuse and fraudulent acts [62]. MM, according to the joint US Commodity Futures Trading Commission, is defined as actions that can artificially shift prices by pressuring them to change [63]. The FBI defines MM as the act of someone or a group trying to influence a free market in their favor [61]. In general, market manipulation refers to any unusual buying or selling committed with the intention of misleading regular shareholders, which affects the market's natural mechanism and investors' decisions, ultimately changing supply-demand dynamics and asset pricing.

Market manipulation is done in two ways. First is *trade-based* manipulation, in which fraudulent traders usually manipulate the market by making fictitious trades with themselves (with no

Table 2. Comparison of FiFrauD and Relevant Approaches

Property	HoloScope [17]	ANTICO [31]	AntiBenford [29]	AnoEdge [57]	FlowScope [30]	Fraudar [56]	SpokEn [58]	CopyCatch [55]	FiFrauD
Market manipulation									✓
Unipartite graph									✓
Generality (directed/attributed)	✓	✓		✓					✓
Spike-aware	✓								✓
Camouflage	✓	✓				✓			✓
Scalability	✓	✓	✓	✓	✓	✓	✓	✓	✓
Accuracy guarantees	✓	✓		✓		✓			✓
Temporal attention	✓	✓							✓
Behavior attention		✓							✓
Structural attention	✓	✓		✓		✓			✓

Table 3. Classification for Financial Fraud

Type of financial fraud	Fraudulent activities
Bank fraud	Mortgage, credit card, money laundering
Insurance fraud	Healthcare, crop, auto
Fraud in financial markets	Securities/stock
Other related financial fraud	Corporate, mass marketing

change in asset ownership) and also by collusive group trades. Second is *order-based* manipulation, in which the traders, using the capacity of strong and fast infrastructures, submit buy/sell orders and quickly delete them before the trade is made, with the aim of inducing a false and fictitious image of the upcoming transaction process. Among the common manipulations, wash trade, prearranged, momentum ignition, marking the close, marking the open, and pump-and-dump are trade-based manipulations, whereas spoofing, layering, and quote stuffing are examples of order-based manipulations. To further study market manipulations and their definitions, please refer to the review paper of [7].

Since trade-based manipulation looks legitimate in appearance and there is no obvious infringement, confronting such frauds is more challenging. Trades appear to be occurring between individuals, but some may be spoofing (without variation in stock possession) or done in collusion and purposeful deception, making them difficult to detect as manipulations [3, 64]. On the other hand, in this type of manipulation, due to the fact that real trades are actually taking place, which are included on the market trading board, ordinary traders are more affected. Therefore, manipulators most often prefer to act based on this type of manipulation to lower the detection probability by bypassing the rules, as they would achieve their goals more easily by deceiving ordinary traders and market supervisors. Accordingly, this study focuses on trade-based manipulations. In the rest of the article, MM refers to this type of manipulation.

MMs are done using various methods. Along with the progress and evolution of the markets, these methods meet new dimensions and are implemented in different forms. In this study, by observing the manipulative behavior in the stock market, analyzing the typical and common methods of manipulation, reviewing the conducted studies, and investigating the opinions of stock market experts, we obtained two findings that are in line with the findings of previous studies.

First, market manipulations are often done in a group and collusive manner. Various reasons can be given for this, among which is that to affect online financial markets and create a fabricated trend in them, lots of trades must typically be made to convince the audience, or regular investors, to adopt a certain position in the trades (targeted by the fraudsters) and engage in specific trading practices. Usually, in these markets, except in rare cases in which atypical volumes are traded, in which manipulations are typically committed by the market's legal entities having lots of stocks,

the likelihood of convincing traders with a few fake trades is low because of the vastness of these types of markets. In other cases, actual individuals typically collude to trick other traders and make fictitious trades by executing multiple orders/trades.

Second, manipulation methods, especially group manipulations, are all done using specific principles and traits despite their variety. That is, every manipulation that is going to occur in the market has at least one or more of these traits and, outside of these traits, it is not possible to cheat and manipulate the market. It is only possible to change the type of use or the combination of these traits in different manipulation behaviors. These main traits of typical and common MM include high intensity, high involvement, temporal spikes, and high cohesiveness, which we discuss in the next section.

4 PROBLEM FORMULATION

In this section, we summarize four key traits of typical manipulations and then formally define our problem.

4.1 Key Traits

Trait 1 (Intensiveness). Fraudsters make many intensive trades with each other within a short time.

Lockstep activities, in which a group of buyers and sellers intensively trade with each other, are most often an important sign of fraud. Fraudulent traders need to make many trades together so that they can influence the asset's supply and demand, mislead other traders by inducing a false and fake atmosphere of the trading process, and push the market's direction in the path that they want. In addition, fraudsters make their transactions in a short period of time to achieve their goal and induce a rapid change in the market situation.

Trait 2 (High Involvement). Fraudsters are more highly involved in trades with each other than with ordinary traders.

Fraudulent traders sometimes make transactions with ordinary traders to camouflage and reduce the risk of detection, but the number of transactions between themselves is much higher than those with others.

Trait 3 (Spikes: Bursts and Drops). Fraudsters conduct their manipulative activity by making a wave of sequential transactions (bursts of activity)—they start trading with each other all at once and stop working suddenly (sudden drops in activity) after reaching the desired goal.

Fraudulent traders pay special attention to the time factor in their manipulation strategies—they try to act in a surging manner in a short period of time to reduce the time cost, then suddenly stop trading.

Trait 4 (High Cohesiveness). Fraudsters conduct manipulative activity in a highly coherent manner. They simultaneously submit orders with similar prices and volumes.

Colluding traders, to ensure that they are trading with each other, plan in such a coherent state that the submission time of orders and the volumes and prices of orders would be similar. As a result, the risk of trading with other traders is reduced, as reported in previous works [7, 10].

Traits 1 and 2 show graph topology, in which fraudsters are highly involved in intensive transactions among themselves and their transactions with others are relatively small. Traits 1 and 3 are related to the temporal aspect, which shows that the manipulative activities are usually accompanied by a wave of surging transactions that starts with sudden bursts and ends with sudden drops after a short period of time. Trait 4 shows the manipulation behaviors in terms of the features of orders. To sum up, manipulative activities within a short period of time create intensive trading streams with sudden bursts and drops in a completely coherent manner that are distinct from common transactions.

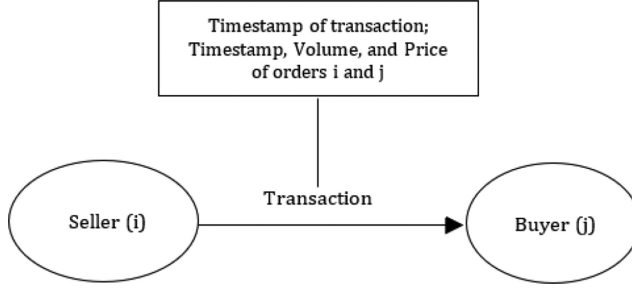


Fig. 1. Graph schema.

In the following sections, we will show that FiFrauD can make comprehensive use of several signals—namely, topology, temporal spikes, and behavior—to spot fraudulent traders that meet the above traits. In FiFrauD, unlike the methods for detecting anomaly patterns in static data, we detect the MM anomaly patterns from a large-scale transactions stream and spot the fraudulent traders.

4.2 Problem Definition

Given a graph stream G_s is the set of an infinite sequence of graphs $G_s = \{G_1, G_2, \dots, G_i, \dots\}$ coming in a stream of time $\{t_1, t_2, \dots, t_i, \dots\}$, where each G_i denotes the graph at the time i . We consider a graph $G_i = (V_i, E_i, X_i)$ as a generic attributed directed unipartite graph that consists of (1) a set of nodes $V_i = \{v_1, v_2, \dots, v_n\}$, where $|V_i| = n$ is the number of nodes; (2) a set of attributed edges $E_i(S, P) = \{e_{ij}(t)\}$, where $|E_i| = m$ is the number of edges; and (3) the edge attribute matrix $X_i = [x_{ij,t}^T] \in R^{m \times p}$, where $x_{ij,t}$ is a p -dimensional attribute vector of the edge between nodes i and j at time t , which includes the transaction time, order submission time, order volume, and order price of nodes i and j . Here, $V = S \cup P$ is composed of x seller (i.e., S) and y purchaser (i.e., P), such that the sets S and P overlap, meaning that a trader can appear both as a seller and a purchaser during the transactions.

G_i is a multi-graph; thus, there can be several edges between two nodes, i.e., a seller can have a transaction with a buyer several times. Nodes V indicate the traders, edges E indicate the transactions, and $e(t)$ details a specific transaction from the seller $i \in S$ to the purchaser $j \in P$ at the time t . G_i 's graph schema is depicted in Figure 1. This study's problem is defined as follows.

Problem 1 (Manipulation detection in transaction streams). Given a stream of directed attributed graphs $\{G_1, G_2, \dots, G_i, \dots\}$, where in each seller–buyer transaction graph G_i , the edge $e_{ij}(t) \in E$ denotes a transaction from the seller i to the buyer j at time t ,

- **Find** a group of suspicious traders, or suspicious transactions in each G_i ,
- **to optimize** a class of suspiciousness metrics under the traits in terms of topology, time, and behavior.

We show that the detection efficiency can be theoretically bounded by optimizing our proposed metric with approximate guarantees. The theoretical analysis is given in Section A.1 of the appendix.

5 PROPOSED METHOD

According to the problem definition, we propose a method to measure the subgraphs' suspiciousness in each G_i , which is shown as G from now on for simplicity in this article. Figure 2 illustrates the proposed framework.

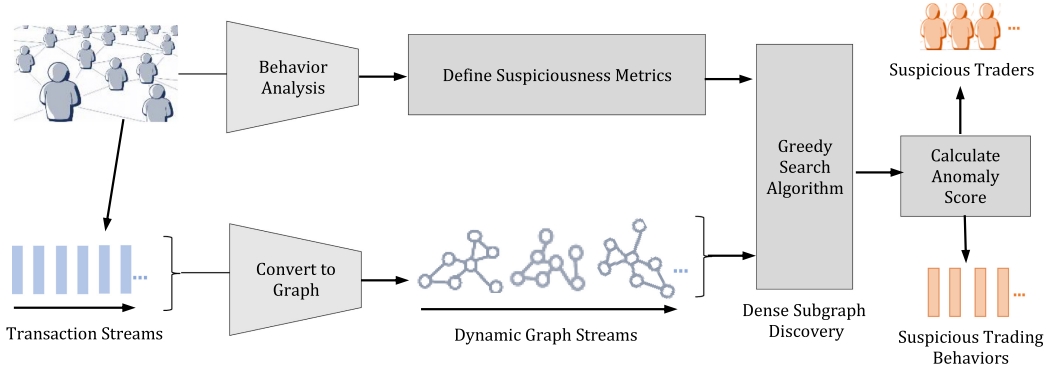


Fig. 2. Visual illustration of FiFrauD framework.

5.1 Metric

The main idea of our approach is to aggregate and mark the edges between node pairs. To provide a formal definition of our approach, we first define $G' = (V, E')$ as a simple transaction graph of $G = (V, E, X)$. For $i \in S$ and $j \in P$, the edge satisfies in the following relationship.

$$e'_{ij} = \begin{cases} 1 & e_{ij}(t) \in E(S, P) \\ 0 & \text{else} \end{cases}, \quad (1)$$

which shows the binary relation between node pairs. Now, we can construct the framework of our metric. Next, we demonstrate how to detect manipulations from three perspectives: topology connection, timestamp, and behavior (Traits 1–4). Finally, we aggregate them for joint use. The symbols used in this article are defined in Table 4.

5.1.1 Overall Framework. Let $g(A \cup B)$ denote a class of metrics for optimization such that the subset of seller nodes $A \subseteq S$ and the subset of purchaser nodes $B \subseteq P$ form the target dense subgraph, which our algorithm is to identify. Based on Trait 1, which says that sellers and buyers do intensive transactions among themselves in the collusion group, density metrics can be appropriate. There are various density metrics, including geometric and arithmetic average degree, that can be used in FiFrauD. However, since the average degree metrics are biased toward the inclusion of a large number of nodes, we propose using density metrics g formed as follows:

$$g(A \cup B) = \frac{1}{|A| + |B|} \sum_{i \in A} \sum_{j \in B} a_{ij} \cdot e'_{ij} \cdot c_{ij} \quad (2)$$

Where e'_{ij} is a binary and defined in Equation (1) and $a_{ij} \geq 0$ is a prior constant whose value can be equal to 1 if there is no additional knowledge about suspiciousness. This constant can be derived as prior information from the degree or extra domain knowledge, such as the time of abnormal behavior. In the next section, we offer a method to set a value to this constant. c_{ij} measures the transaction between seller i and buyer j . It is known as *edge suspiciousness* in G' and is estimated as follows:

$$c_{ij} = \sum_{e_{ij}(t) \in E} b_{ij}(t), \quad (3)$$

which is aggregated with the suspiciousness of each transaction between i and j . *Transaction suspiciousness* $b_{ij}(t)$ is defined as follows:

$$b_{ij}(t) := P(i \in A) \cdot P(j \in B) \cdot P(i \in A, j \in B | t), \quad (4)$$

Table 4. Symbols and Definitions

Symbol	Definition
S	Sellers
P	Purchasers
G_i	Attributed directed unipartite multi-graph, $G = (V, E, X)$
V	Nodes of graph G , $V = S \cup B$
E	Transaction edges of graph G , $E = \{e_{ij}(t)\}$
$e_{ij}(t)$	Specific transaction from user i to merchant j at time t , its value is equal to 1.
X	The edge attribute matrix includes transaction time and submission time, volume, and price of order of nodes i and j , $X = [x_{ij,t}^t] \in \mathbb{R}^{ m \times p}$
A	Subset of sellers
B	Subset of purchasers
G'	Directed unipartite graph, $G' = (V, E')$
E'	Binary transaction relations, $E' = \{e'_{ij}\}$
e'_{ij}	Binary relation between seller i and buyer j
$g(A \cup B)$	Density metric
T	Transaction time cycles, $T = \{T^{(1)}, \dots, T^{(l)}\}$
$T^{(k)}$	Transaction time slices, $T^{(k)} = \{T_1^{(k)}, \dots, T_d^{(k)}\}$
$\mathcal{M}$	Group of node suspiciousness metrics, $\mathcal{M} = \{\mathfrak{T}, \mathfrak{C}, \mathfrak{S}\}$
C	Edge suspiciousness metric
$F(\cdot)$	Transition function
$K(\cdot)$	Joint probability function

where $P(i \in A)$ and $P(j \in B)$ are the prior estimate on the probability of a seller node i , and a buyer node j being part of the suspicious seller set A , and the suspicious buyer set B , respectively. $P(i \in A, j \in B | t)$ is the conditional probability of a transaction being manipulated at time t . This means that for identifying dense blocks, we measure transaction suspiciousness from three aspects: the suspiciousness of (1) the seller, (2) the buyer, and (3) the transaction itself.

In the following sections, by using this overall framework for our proposed metric $g(A \cup B)$, we show how to satisfy the remaining traits. To do this, we define node suspiciousness $P(i \in A)$ and edge suspiciousness $P(i \in A, j \in B | t)$ in a way that different edge attributes are considered. This makes it possible to be more accurate in detecting blocks in low-density conditions. In general, some of the main advantages of metrics of this form are that they can follow some basic properties mentioned in [28]. Also, the optimization is scalable and with theoretical guarantees.

5.1.2 Prior Estimate. Later on, we present how to approximate $P(i \in A)$ and $P(j \in B)$ according to the above traits (see Section 4.1). Due to the similarity of how we approximate these two likelihoods, we will only discuss how to approximate $P(i \in A)$ in order to avoid repetition.

High Intensiveness. Fraudsters tend to make intensive transactions with each other in a short period of time (Trait 1). That is, manipulative activities create large transaction streams having adjoining timestamps. Heterogeneous temporal patterns can be revealed using different methods. We use the Gini Index, designed to capture wealth inequality within a state. In our approach, we examine how the number of transactions (as wealth) is distributed across time slices (as people). The practical data range for Gini is $[0, 1]$.

Suppose that the k -th time period $T^{(k)}$ is split into d time slices, which are in the form of $\{T_1^{(k)}, T_2^{(k)}, \dots, T_d^{(k)}\}$. Let $\{x_1^{(i)}, x_2^{(i)}, \dots, x_d^{(i)}\}$ be indexed in ascending sort, where $x^{(i)}$ is the aggregation of the number of transactions within a time slice given trader i . Trait 1 is formulated as

follows:

$$\mathbb{T}_i \left(T^{(k)} \right) = \frac{2 \sum_{j=1}^d j \cdot X_j^{(i)}}{d \sum_{j=1}^d X_j^{(i)}} - \frac{d+1}{d}. \quad (5)$$

Then, we have that $P(i \in A) \propto F(\mathbb{T}_i)$, where $F(\cdot)$ transits metrics into probability estimation. We consider $T^{(k)}$ equal to one trading day and divide it into seven 30-minute slices of time.

High Involvement. The colluding traders are more suspicious if most of their transactions are made with suspicious traders and they rarely trade with ordinary traders (Trait 2). Mathematically, for $t \in T_s^{(k)}$, this trait is captured as follows:

$$P(i \in A) \propto F(\zeta_i), \text{ where } \zeta_i \left(T^{(k)} \right) = \frac{\sum_{\tau \in T_s^{(k)}} e'_{i*} \cdot a_{ij} + \sum_{\tau \in T_s^{(k)}} e'_{*i} \cdot a_{ij}}{\sum_{\tau \in T^{(k)}} e'_{i*} \cdot a_{ij} + \sum_{\tau \in T^{(k)}} e'_{*i} \cdot a_{ij}}. \quad (6)$$

Within the time slice $T_s^{(k)}$, $\sum_{\tau \in T_s^{(k)}} e'_{i*}$ and $\sum_{\tau \in T_s^{(k)}} e'_{*i}$ are the in-degree and out-degree, respectively, of node i . The edges are weighted with the constant a_{ij} . $\zeta_i \in [0, 1]$ quantifies the involvement of node i during the manipulation period compared with other periods. In fact, ζ_i dynamically regulates node weights and also penalizes nodes with few transactions within the group.

Spikes (Temporal Bursts and Drops). Fraudsters tend to make their trades successional and suddenly to reduce the time cost such that this abrupt surge of trades results in sudden bursts accompanied by sudden drops one or more times (Trait 3).

Concerning this trait, there may be a significant difference between the suspiciousness rates of two different groups of traders even if these two groups act identically based on Traits 1 and 2 because the difference between these two groups cannot be identified only by the graph's connections. Thus, in the FiFrauD framework, we also consider the temporal attribute (these sudden bursts and drops) to detect suspicious groups.

In order to consider the temporal attribute, inspired by [17], we have indicated the list of timestamps of (input/output) edges connected to traders' nodes by $\mathcal{F}_i$ and $\mathcal{F}_j$, respectively, for sellers i and buyers j . To simplify the notation, for a given node, we use $\mathcal{F}$ without any subscript. Consider $\Gamma = \{(t_0, c_0), (t_1, c_1), \dots, (t_e, c_e)\}$ as the time series of $\mathcal{F}$; that is, the histogram of $\mathcal{F}$. c_i value shows the number of timestamps in the time bin $[t_i - \Delta t/2, t_i + \Delta t/2]$, with bin size Δt . The bin size of the histogram is calculated based on the maximum of Sturges criteria [65] and the Freedman-Diaconis criteria [66], whose value is set in the experiments.

To account for the burst and drop patterns in Trait 3, we should locate the burst's starting point and the drop's ending point within the time series Γ . Assume that the point (t_m, c_m) with the highest c_m value is the burst point. The awakening point of a burst, or the starting point, can be identified by drawing a straight line from the beginning to the burst point. The time series Γ (red polygonal line) of transactions, the awakening point (t_a, c_a) for the burst point (t_m, c_m) , and the ancillary straight line l (black dotted line) connecting (t_0, c_0) to (t_m, c_m) are shown in Figure 3. The point on the time series Γ where the distance to l is maximized is the burst's awakening point (t_a, c_a) . According to the figure, the awakening point, given the line perpendicular to line l , fulfills Equation (7).

$$t_a = \arg \max_{(c, t) \in \Gamma, t < t_m} \frac{|(c_m - c_0)t - (t_m - t_0)c + t_m c_0 - c_m t_0|}{\sqrt{(c_m - c_0)^2 + (t_m - t_0)^2}}. \quad (7)$$

Since there may be multiple bursts, it is not sufficient to identify the awakening point for just one. Therefore, the sub-burst points and their related awakening points also need to be found.

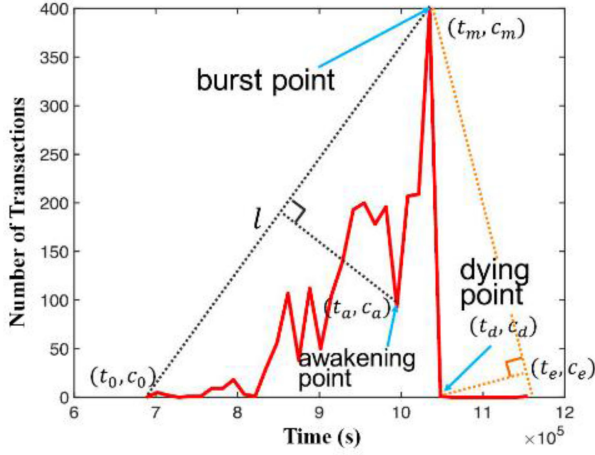


Fig. 3. Detection of burst and drop.

Algorithm 1 summarizes the *MultiBurst* algorithm that identifies more bursts by removing the current burst period (t_a, t_m) and recursively searching the remaining time series on the left and right until none are left. The algorithm outputs a list of bursts, including awakening and burst times, altitude differences, and slopes.

Therefore, the nodes' suspiciousness with awareness of the bursts satisfies $P(i \in A) \propto F(\zeta_i)$ such that ζ_i is defined as the involvement ratio of node i in multiple bursts.

$$\zeta_i \left(T^{(k)} \right) = \frac{\zeta \left(T_s^{(k)} \right)}{\zeta \left(T^{(k)} \right)}, \text{ and } \zeta(T) = \sum_{(t_a, t_m)} \Delta c_{am} \cdot s_{am} \sum_{t \in T} 1(t \in [t_a, t_m]) \quad (8)$$

s_{am} , representing the slope of the MultiBurst output, is employed as the weight proportional to the current burst's intensity. It is worth noting that $\zeta_i \in [0, 1]$ measures how much a trader is involved in multiple bursts during the manipulation period. Considering the FiFrauD framework, only those traders who participate simultaneously in intensive bursts can be identified as fraudsters; therefore, Trait 3 is fulfilled. In other words, honest traders who occasionally get involved in bursts will not be detected as FP cases.

ALGORITHM 1: *MultiBurst*

Input Time series Γ of node v , beginning index i , end index j

Output A list of awakening-burst point pairs

s_{am} : slope of the line passing through each point pair

Δc_{am} : altitude difference of each point pair.

If $j - i < 1$, **then**

return

(t_m, c_m) = point of maximum altitude in Γ_{ij} .

(t_a, c_a) = find awakening point as Equation (7) in Γ_{ij} .

$\Delta c_{am} = c_m - c_a$, and $s_{am} = \Delta c_{am} / (t_m - t_a)$

Append $\{(t_a, c_a), (t_m, c_m)\}$, s_{am} , and Δc_{am} into the output.

MultiBurst($\Gamma, i, a - 1$)

k = Find the first local min position from indices $m + 1$ to j

MultiBurst(Γ, k, j)

Sudden drops are another prevalent characteristic of manipulation behaviors according to Trait 3, as fraudsters typically end their activity abruptly after completing their scheme. By defining the dying point as the end of a sudden drop, we can use this suspicious pattern. In this regard, a second ancillary straight line is created from (t_m, c_m) to (t_e, c_e) , as shown in Figure 3. The sudden drop can be found by identifying the dying point (t_d, c_d) through the maximization of the distance to the straight line and calculating the absolute slope value $s_{bd} = (c_m - c_d)/(t_d - t_m)$ between the burst and dying points. As time series Γ may have multiple drops, Algorithm 2 is used to identify the drop with the highest fall.

ALGORITHM 2: *MaxDrop*

Input Time series Γ of node v , beginning index i , end index j

Output Δc_{bd} : altitude difference of the point pair

s_{bd} : slope of the line passing through the point pair.

If $j - i < 1$ **then**

 return

$(t_m, c_m) =$ point of maximum altitude in Γ_{ij} .

$(t_d, c_d) =$ find dying point in Γ_{ij} by definition.

$\Delta c_{bd} = c_m - c_d$, and $s_{bd} = \Delta c_{bd}/(t_d - t_m)$

If Δc_{bd} is larger than the drop found in the output **then**

 Overwrite output with Δc_{bd} and s_{bd} .

MaxDrop ($\Gamma, i, m-1$)

MaxDrop (Γ, d, j)

To detect the next potential drops, the *MaxDrop* algorithm removes the current drop period (t_m, t_d) and recursively searches the remaining time series on the left and right until none are left. After detecting the drops, the highest drop fall Δc_{bd} and its drop slope s_{bd} are returned by the *MaxDrop* algorithm. A sudden drop in the created edges between nodes is unusual and suggests that the nodes are suspicious. For this reason, we incorporate the weighted drop slope into Equation (2) as additional knowledge for suspiciousness.

$$a = \Delta c_{bd} \cdot s_{bd} \quad (9)$$

For simplicity, the edges' subscripts are omitted for each edge connected to node i . Note that to increase or decrease the previous suspiciousness, it is essential to employ the constant a_{ij} .

According to Theorem 3, with regard to the proposed approach for the detection of burst and drop, Trait 3 creates a temporal barrier for fraudsters. The theorem with its proof is given in the theoretical analysis in Section A.1 of the appendix.

Fusion. The next issue is how to aggregate metrics for $P(i \in A)$, which is necessary to fully use different signals. To aggregate metrics, the following requirements should be met: (a) high suspiciousness should be in a saturated regime, (b) the output should be in the form of probability and non-zero, and (c) it should be scalable for more metrics.

An intuitive way is the direct multiplication of metrics. However, since small variables may lead to the problem of the disappearance of suspiciousness, this method seems inefficient. In this article, we use a sigmoid function for better probability estimation. Assume that $\mathcal{M}$ is a set of metrics, where $\mathcal{M} = \{\mathbb{F}, \mathbb{C}, \mathbb{S}\}$. Given trader i , $F(\cdot)$ is defined as follows:

$$P(i \in A) \propto F(\mathcal{M}_i), \text{ where } F(\mathcal{M}_i) = \frac{1}{1 + b^{(1-2K(\mathcal{M}_i))}}, \quad (10)$$

where the joint probability K is

$$K(\mathcal{M}_i) = \sum_{m_{ij} \in \mathcal{M}_i} \gamma_j \cdot m_{ij}. \quad (11)$$

For constants $\gamma_j > 0$, we have that $\sum \gamma_j = 1$. The constants can be set to different values (their default value is $\frac{1}{M}$). $F(\cdot)$ contains a sigmoid curve, which is limited by the probability bounds. The saturation interval can be set as $b > 1$. Moreover, in a real-world environment, fraudulent traders may only behave suspiciously from some perspectives, yet we can have a high probability of suspiciousness using the saturation function. That is, the value of one signal is not removed despite the very small value of other signals. For example, a suspicious burst and drop can still maintain a high suspiciousness score in the presence of low scores on other metrics at a low fraudulent density. Therefore, in most cases, FiFrauD guarantees noise-anti.

5.1.3 Edge Suspiciousness. The approximation of $P(i \in A)$ and $P(j \in B)$, or the nodes' suspiciousness, has been obtained thus far. The next step involves determining how to assess each transaction, with the main goal of approximating its suspiciousness based on Trait 4.

High Cohesiveness. Fraudsters in a suspicious transaction are coordinated and coherent in terms of (a) order submission time, (b) order volume, and (c) order price. Consider that seller i and buyer j submit their order at time t_i and t_j with volume v_i and v_j and price p_i and p_j . The transaction suspiciousness between them, i.e., $P(i \in A, j \in B|t)$, is estimated as follows:

$$P(i \in A, j \in B|t) \propto F(C_{ij}), \text{ where } C_{ij} = \begin{cases} 0 & |t_i - t_j| > \mathcal{T}_t \vee |P_i - P_j| > \mathcal{T}_p \\ \zeta_{ij} & \text{else} \end{cases}, \quad (12)$$

where the collusiveness ζ_{ij} is

$$\zeta_{ij} = a_{ij} \left[\alpha \left(1 - \frac{|t_i - t_j|}{\mathcal{T}_t} \right) + \beta \left(1 - \frac{|p_i - p_j|}{\mathcal{T}_p} \right) + \gamma \left(\min \left(\frac{v_i}{v_j}, \frac{v_j}{v_i} \right) \right) \right] \quad (13)$$

Where a_{ij} is the prior constant (see Equation (9)). Coefficients α , β , and γ determine the relative significance of time, price, and volume, respectively ($0 \leq \alpha, \beta, \gamma \leq 1$, and $\alpha + \beta + \gamma = 1$). Since simultaneity indicates more collusion than coherence in price and volume, and price also indicates more collusion than volume, we set $\alpha > \beta > \gamma$. $\mathcal{T}_t$ and $\mathcal{T}_p$ as the model parameters. In a transaction, if the time difference between the order submission of seller i and buyer j is beyond $\mathcal{T}_t$ (in seconds), or their price deviates beyond $\mathcal{T}_p$ (in Rial Currency), we do not consider this transaction. The main point is that ζ_{ij} shows the collusiveness between the two traders i and j such that when the coherence between them is greater, we assign a higher suspiciousness to each transaction; As a result, Trait 4 is satisfied. Our algorithm achieves the best results with $\alpha = 0.5$, $\beta = 0.35$, $\gamma = 0.15$, $\mathcal{T}_t = 180$, and $\mathcal{T}_p = 10$.

In summary, FiFrauD can improve the manipulation detection accuracy in dynamic graphs (even with low fraudulent density) by using a class of metrics for the suspiciousness of nodes and edges in a dynamic and camouflage-resistant manner.

5.2 Algorithm

In this section, how to optimize g is described. FiFrauD iteratively finds and then eliminates a single node having the lowest score based on Charikar's greedy peeling method [67], as nodes with lower scores are less suspect. To do so, we start by defining the score given to each node i , which must homologize the elements of g . To achieve this, we take into account the node information in G' , i.e., the degree of nodes. This score is defined as Equation (14):

$$w_i(S) = \begin{cases} \sum_{j \in B} a_{ij} \cdot e'_{ij} \cdot c_{ij} & i \in A \\ \sum_{j \in A} a_{ji} \cdot e'_{ji} \cdot c_{ji} & i \in B \\ \sum_{j \in B} a_{ij} \cdot e_{ij} \cdot c_{ij} + \sum_{j \in A} a_{ji} \cdot e'_{ji} \cdot c_{ji} & i \in A \wedge i \in B \end{cases} \quad (14)$$

where $S = A \cup B$. To efficiently find the node with the lowest score, we construct a priority tree. The priority tree dynamically updates the score of other nodes when a node is removed, ensuring

that the new minimum is maintained as priorities change. Let $S^{(k)}$ be the subgraph's nodes in the k -th iteration, and let $S^{(0)}$ be the initial set of nodes $S \cup P$. This algorithm includes a greedy procedure of peeling nodes from the graph. Specifically, the k -th iteration is to find a local optimum g , which is evaluated over the rest of the nodes $S^{(k)} = S^{(k-1)} \setminus \{i\}$ and removes node i with the lowest score. The iterations generate a diminishing sequence of subsets of nodes $S^{(0)} \supset S^{(1)} \supset \dots \supset S^{(|S|+|P|)}$ until the rest are empty. Then, the algorithm outputs the densest subgraph with the maximum g and the subset of nodes S^* as the final result. In a precise manner, the output includes the score of the identified subgraphs, the nodes and edges of these subgraphs, and the scores of these nodes and edges. This entire procedure is summarized in Algorithm 3. Additionally, the subsequent subgraph can be intuitively found by removing the edges from the current subgraph and repeating the above procedure. Recall that Algorithms 1 and 2 are used to calculate Equation (14).

ALGORITHM 3: *FiFrauD*

Input Graph Stream $G_S = \{G_1, G_2, \dots, G_t, \dots\}$, each graph as $G_t(S \cup P, E, X)$

Output Suspicious trader set $S^* \subset (S \cup P)$

while not end of stream **do**

$S^* \leftarrow \emptyset$

$S^{(0)} \leftarrow S \cup P$

$\{w_i\} \leftarrow$ compute all node scores by Equation (14)

$P^{(0)} \leftarrow$ construct priority tree from $S \cup P$

while $k = 1, \dots, |S| + |P|$ **do**

$i \leftarrow$ find the minimum node from $P^{(k-1)}$

$S^{(k)} \leftarrow S^{(k-1)} \setminus \{i\}$

$\{w_i\} \leftarrow \forall_j$ if e_{ij} or e_{ji} then recalculate node score

$P^{(k)} \leftarrow$ assign ∞ to leaf i then recalculate tree

if $g(S^{(k)}) > g(S^*)$ **then**

$S^* \leftarrow S^{(k)}$

return S^* that maximizes g in Equation (2)

6 EXPERIMENTS

In this section, we design the experiments to answer the following questions:

- (1) **Q1. Effectiveness for pattern detection:** How accurately does FiFrauD detect the expected behavioral patterns (P1–P4) of traders for manipulation? Does FiFrauD have an advantage over other baselines?
- (2) **Q2. FiFrauD spots real-world fraudsters:** What patterns does FiFrauD detect in real-world datasets? How about the behavior of the suspicious traders?
- (3) **Q3. Scalability:** Does our method scale linearly with the number of edges?

6.1 Real-World Datasets

We use the real-world datasets of the Iranian Stock Market to evaluate the proposed framework. The statistical information of the datasets is listed in Table 5. As shown in the table, these data are related to three different stocks that are anonymized by using distinguishable IDs, such as STK1, STK2, and STK3, due to the confidentiality of the data. We use the first five datasets to evaluate the methods. The dataset related to each stock includes all the transaction records of that stock in 1 month. We use the last dataset to analyze the time complexity of our algorithm (the second part of Table 5).

To better understand the data, an excerpt of complete information of the genuine data STK1_Dec2021 is shown in Table 6. The dataset contains 13 fields, showing a complete overview of

Table 5. Statistics of Real-World Datasets

Dataset	# Nodes	# Edges	# Sellers	# Buyers	# Transactions	Time Span	# True label
STK1_Dec2021	34,840	80,610	25,856	18,374	90,430	Dec. 01–Dec. 29, 2021	10
STK3_Dec2021	74,575	203,270	54,950	43,964	226,569	Dec. 08–Dec. 29, 2021	9
STK2_Dec2021	46,064	113,862	34,074	25,417	129,496	Dec. 08–Dec. 29, 2021	9
STK1_Jan2022	25,436	54,264	17,918	12,267	62,738	Jan. 01–Jan. 31, 2022	4
STK3_Jan2022	70,764	216,558	56,807	40,278	240,969	Jan. 01–Jan. 31, 2022	4
STK3_6 Month	118,230	1,040,809	86,224	71,256	1,161,176	Oct. 02, 2021–Mar.1 9, 2022	-

Table 6. An Excerpt of Complete Information of The STK1_Dec2021 Data

TRD_NR	TRD_DATE	TRD_TM	TRD_PR	TRD_QTY	CLIENT_SO	CLIENT_BO	SO_ORD_ENTRY_DTTM	BO_ORD_ENTRY_DTTM	SO_PR	BO_PR	SO_QTY	BO_QTY
153	2021-12-01	09:00:48	1890	48937	272647	282239	2021-12-01 09:00:38	2021-12-01 09:00:48	1890	1890	53305	111457
1224	2021-12-01	09:12:27	1925	4000	235093	177530	2021-12-01 09:12:27	2021-12-01 09:12:22	1765	1925	330608	4000
3123	2021-12-01	09:46:53	1879	51000	594188	566297	2021-12-01 09:46:53	2021-12-01 09:45:00	1879	1879	55104	51000
...	...	...	...	...	...	...	...	...	...	...	...	...

Table 7. Definition of Fields of Datasets

Field	Definition	Field	Definition
TRD_NR	Transaction ID	SO_ORD_ENTRY_DTTM	Date and time of the seller's order entry
TRD_DATE	Transaction date	BO_ORD_ENTRY_DTTM	Date and time of the buyer's order entry
TRD_TM	Transaction timestamp	SO_PR	Seller's order price
TRD_PR	Transaction price	BO_PR	Buyer's order price
TRD_QTY	Transaction volume	SO_QTY	Seller's order volume
CLIENT_SO	Seller ID	BO_QTY	Buyer's order volume
CLIENT_BO	Buyer ID		

every transaction. Due to investor confidentiality, the buyer and seller information (counterparty codes) are anonymized with distinguishable IDs such as 282239, 177530, and 566297. The dataset fields are defined in Table 7.

There are real labels for the first five datasets. Regulatory experts at the **Tehran Stock Exchange (TSE)** identified suspicious traders and provided us with their IDs. They identified the fraudulent traders based on behaviors such as trades with a large and unusual volume of shares among a group, buying and selling with no change in ownership of shares, consecutive and many trades to influence the market trend, sharp rise or fall in share prices in short-term trading to induce the formation of a specific trend in the market, and quick execution of suspicious trades in the first or last trading day with the aim of misleading ordinary traders from predicting the trading trend. In Section 6.3.2, we examine whether our proposed method can detect these labels reported by the TSE. The number of suspicious traders reported by the TSE in each dataset is listed in Table 5. It should be noted that the proposed method is an unsupervised approach and the labeled dataset is only used for performance evaluation.

6.2 Experiment Setup

Baselines: We select the following methods as baselines: AntiBenford [29], ANTICO [31], AnoEdge [57], Fraudar [56], and HoloScope [17], which are unsupervised graph-based methods that detect anomalies by finding dense subgraphs.

Evaluation Metric: We compare FiFrauD with other baselines using AUC, F-measure, precision, recall, and accuracy to evaluate its performance in detecting suspicious traders.

Hyperparameter settings: In the proposed model, after performing a grid search to tune the hyperparameters, finally, the number of blocks for detection is set to 2, the size of b in the fusion function is set to 10, and the size of the time window is set to 1 day because manipulation behaviors mainly target the intraday transactions for taking profit [7, 10]. In all experiments, only significant multiple bursts are considered for the time series of nodes, and awakening-burst point pairs are kept only if their altitude difference Δc is greater than half of the largest altitude difference in the time series. In all of the baselines, experiments were done on different values of the number of blocks, and the best results were reported. Other baseline hyperparameters are set to their default values.

Experiment environment: All experiments are performed on a 2.3 GHZ Intel Core i7-10510U CPUs processor and 16 GB RAM running Windows. The data are preprocessed and prepared in ORACLE SQL Developer 19.2. All detection algorithms are implemented in Python 3.9., and a part of the AntiBenford algorithm (DSP codes) is implemented in C++. All the baselines are implemented based on their publicly available codes.

6.3 Experimental Results

6.3.1 Q1. Effectiveness in Pattern Detection. To verify the performance of FiFrauD to detect different types of anomalies, we inject various schema into dataset STK1_Dec2021 by considering the various behavior patterns in the manipulation scenario summarized in Section 4.1. Based on the four behavior patterns P1 to P4, we run five experiments on real data with injected labels. The first four experiments are performed by injecting each of the patterns and traits separately, and the last experiment is performed by injecting all the patterns together.

In these experiments, for injection, clean data STK1_Dec2021, meaning all suspicious traders identified by FiFrauD are eliminated, is used as the background. Then, the characteristics of the fraudsters' behaviors are generated synthetically based on behavioral patterns and injected into the clean data while maintaining statistical properties such as mean and variance. It must be noted that due to the fraudulent traders' attempt to behave normally, there are camouflage edges in real and also synthetically generated data. For testing on various fraudulent densities, the number of fraudulent traders ranges from 200 to 2,000, resulting in fraudulent densities ranging from 0.01 to 1.0 for all experiments (Figures 4(a)–4(e)). These traders, in a random manner, are sellers, buyers, or both.

Figures 4(a) to 4(e) demonstrate the performance of FiFrauD and baselines for detecting different patterns in real data with injected labels. We report the performance results of each trait for precise comparisons. As we can see, FiFrauD in Figure 4(a) consistently outperforms all baseline methods for the P1 pattern by achieving 62% improvement in the lowest density and 42% in the highest density. Among the baselines, ANTICO and AnoEdge have higher accuracy than the others, which indicates that they pay attention to the trait of high intensiveness, while AntiBenford gives the worst accuracy.

Figure 4(b) shows the performance for detecting the P2 behavior pattern. As seen, ANTICO and HoloScope also achieve high accuracy (marked by triangle and rhombus, respectively) because these methods, in their algorithm, paid some attention to this trait. However, both methods, especially HoloScope, give low accuracy for low-density cases. Figure 4(c) demonstrates the performance of the methods for detecting surging and sudden manipulations. HoloScope has high accuracy even in low fraudulent density since its purpose is mainly to identify such frauds. However, FiFrauD, with an accuracy difference of 8%, surpasses it as well. According to Figure 4(d),

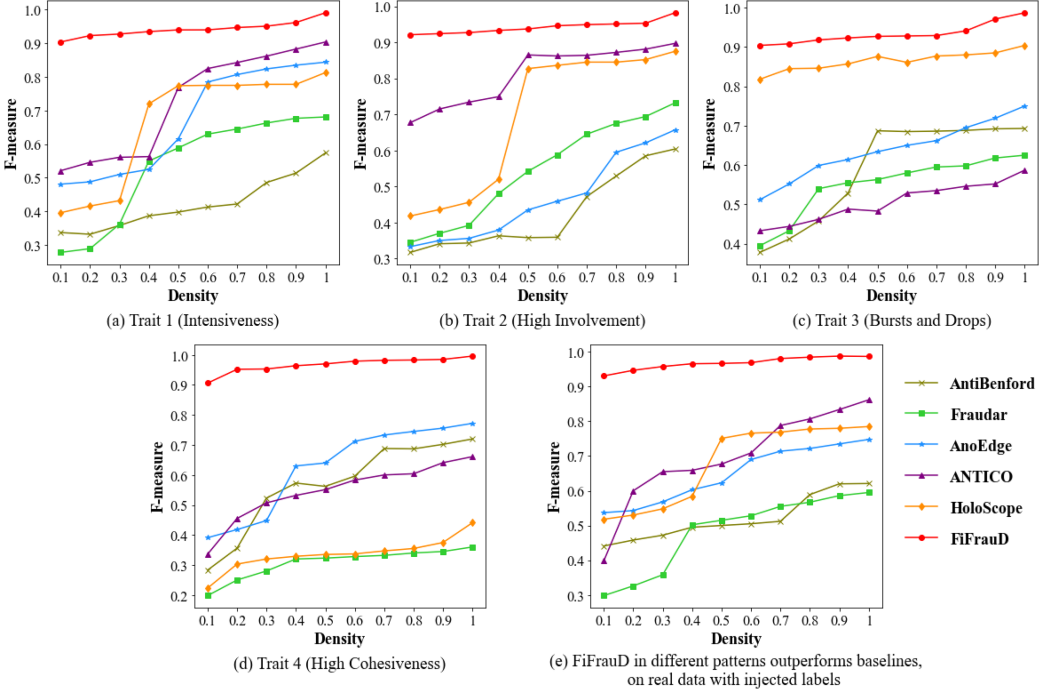


Fig. 4. Comparison of performance on real data with injected labels in F-measure. FiFraud outperforms other baselines under different density settings with (a) injection of the P1 behavior pattern, (b) injection of the P2 behavior pattern, (c) injection of the P3 behavior pattern, (d) injection of the P4 behavior pattern, and (e) injection of all behavior patterns P1 to P4 together. The horizontal axis shows the fraudulent density, i.e., the proportion of the number of injected fraudulent traders over the total number of fraudsters.

FiFraud consistently outperforms all baseline methods for the P4 pattern, achieving 71% improvement in the lowest density and 64% in the highest density.

Figure 4(e) shows the performance evaluation results of FiFraud against different numbers of injected traders (with different injection density) based on all four manipulative behavior patterns. FiFraud obtains the best F-measure and displays clear advantages over the baselines. Fraudsters with low density are much more difficult to detect than those with high density. Hence, better methods can accurately detect them. When the fraudulent density decreases from right to left along the horizontal axis, FiFraud can detect fraudulent density as low as 0.1 with a high F-measure, whose accuracy is much better than 54%, which belongs to the best baseline. This issue is also true in each of the patterns (Figures 4(a)–4(d)). In addition, FiFraud, using several signals, has a high accuracy in detecting fraudulent traders (ideally about 99.6% of accuracy). As a result, it leads to a significant improvement in detection.

In Tables 8 and 9, the performance of FiFraud and other baselines, respectively, for the highest (1) and lowest (0.1) fraudulent density, are reported based on AUC, precision, recall, and accuracy metrics. According to the table, FiFraud, based on all metrics, achieves the best results among the baselines. As can be seen, FiFraud at a density of 0.1 (i.e., when there are only 200 fraudulent traders) has an F-measure and AUC of 90%, a precision of 94%, and the recall and accuracy of 99%, which is better than all competitors. This means that FiFraud, even when the number of fraudsters is very small, can detect all of them with minimal false alarms, i.e., low FP. Furthermore, the AUC score shows that the probability that FiFraud assigns a larger probability to a random fraudster

Table 8. Experimental Results on Real Data with Injected Labels for the Highest Fraudulent Density

Pattern	Metrics	AntiBenford	Fraudar	AnoEdge	ANTICO	HoloScope	FiFrauD
P1	AUC	0.561	0.371	0.852	0.856	0.588	0.972
	Precision	0.433	0.802	0.924	0.921	0.860	0.989
	Recall	0.873	0.592	0.777	0.888	0.772	0.994
	Accuracy	0.981	0.970	0.991	0.990	0.974	0.998
P2	AUC	0.467	0.790	0.475	0.888	0.540	0.888
	Precision	0.465	0.793	0.852	0.960	0.821	0.970
	Recall	0.860	0.680	0.534	0.847	0.936	0.996
	Accuracy	0.976	0.783	0.966	0.994	0.981	0.994
P3	AUC	0.462	0.732	0.823	0.891	0.676	0.920
	Precision	0.574	0.818	0.690	0.550	0.850	0.982
	Recall	0.744	0.506	0.830	0.629	0.965	0.993
	Accuracy	0.958	0.984	0.952	0.958	0.964	0.98
P4	AUC	0.577	0.771	0.847	0.806	0.423	0.895
	Precision	0.71	0.80	0.880	0.794	0.544	0.993
	Recall	0.74	0.236	0.688	0.576	0.371	0.999
	Accuracy	0.978	0.956	0.972	0.979	0.969	0.979
P1-P4	AUC	0.677	0.48	0.539	0.893	0.552	0.936
	Precision	0.582	0.755	0.854	0.877	0.990	0.998
	Recall	0.666	0.490	0.666	0.847	0.650	0.977
	Accuracy	0.984	0.917	0.945	0.980	0.964	0.988

Table 9. Experimental Results on Real Data with Injected Labels for the Lowest Fraudulent Density

Pattern	Metrics	AntiBenford	Fraudar	AnoEdge	ANTICO	HoloScope	FiFrauD
P1	AUC	0.381	0.447	0.463	0.807	0.518	0.856
	Precision	0.396	0.396	0.442	0.511	0.337	0.887
	Recall	0.294	0.214	0.527	0.533	0.480	0.926
	Accuracy	0.973	0.945	0.962	0.940	0.944	0.990
P2	AUC	0.454	0.43	0.439	0.873	0.527	0.841
	Precision	0.243	0.278	0.271	0.875	0.512	0.864
	Recall	0.455	0.455	0.433	0.553	0.353	0.990
	Accuracy	0.973	0.976	0.960	0.982	0.966	0.991
P3	AUC	0.471	0.480	0.514	0.830	0.611	0.856
	Precision	0.524	0.544	0.508	0.554	0.798	0.887
	Recall	0.296	0.310	0.515	0.355	0.840	0.926
	Accuracy	0.911	0.940	0.992	0.956	0.957	0.967
P4	AUC	0.555	0.469	0.482	0.787	0.446	0.840
	Precision	0.214	0.160	0.321	0.266	0.146	0.920
	Recall	0.418	0.250	0.499	0.455	0.470	0.995
	Accuracy	0.958	0.962	0.943	0.976	0.974	0.978
P1-P4	AUC	0.489	0.468	0.503	0.812	0.430	0.901
	Precision	0.524	0.492	0.70	0.499	0.889	0.944
	Recall	0.381	0.214	0.435	0.331	0.366	0.920
	Accuracy	0.962	0.958	0.977	0.943	0.945	0.984

Table 10. Experimental Results on Real Data with Real Labels

Methods	STK1_Dec2021		STK3_Dec2021		STK2_Dec2021		STK3_Jan2022		STK1_Jan2022	
	AUC	Recall	AUC	Recall	AUC	Recall	AUC	Recall	AUC	Recall
AntiBenford	0.577	0.8	0.487	0.777	0.657	0.666	0.695	0.2	0.305	0.666
Fraudar	0.472	0.55	0.539	0.433	0.381	0.4	0.522	0.2	0.386	0.5
AnoEdge	0.340	0.8	0.480	0.55	0.443	0.45	0.552	0.2	0.422	0.6
ANTICO	0.826	0.8	0.763	0.777	0.845	0.666	0.653	0.4	0.848	0.8
HoloScope	0.386	0.7	0.665	0.333	0.448	0.555	0.676	0.285	0.460	0.666
FiFrauD	0.895	1.0	0.945	1.0	0.896	0.9	0.821	0.8	0.898	1.0

than to a random normal trader is at least 90%, which is considered excellent performance. At the highest density (i.e., when there are 2,000 fraudulent traders), the minimum recall is 99%, and the minimum precision is 97%; that is, FiFrauD has been able to identify almost all suspicious traders with minimal errors.

Finally, in summary, according to the results presented in Figures 4(a) to 4(e) and Tables 8 and 9, it can be clearly seen that FiFrauD, due to its comprehensive use of different signals in its detection framework, has an accurate performance even in a very low fraudulent density. The better a method is, the lower the density it can detect well; FiFrauD again performs well in the situation in which the number of fraudsters is very small, and it can detect almost all the fraudsters with the lowest FP and highest probability. All the baselines performed poorly in low density, and even in the highest density, Fraudar and HoloScope can only detect 49% and 65% of the fraudsters, respectively, yet not with an appropriate AUC score.

6.3.2 Q2. Spotting Real-World Fraudsters.

6.3.2.1 Analysis of FiFrauD. We apply FiFrauD to five stock market datasets with different stocks and time spans (see Table 5) to spot suspicious real-world traders and analyze their behavior patterns. The suspicious traders were prosecuted and reported by the TSE experts, whose number for each dataset is given in Table 5. We assign each trader a binary label, in which honest traders are labeled as 0 and fraudulent traders as 1. Here, we report the performance of methods through AUC and recall, widely used in stock-market risk management. AUC is appropriate for evaluating the performance of imbalanced data and establishes a relative trade-off between benefits and costs; recall checks what percentage of suspicious real-world traders were detected by the algorithms.

According to Table 10, FiFrauD has achieved higher scores than other baselines in spotting fraudulent traders in all five datasets. FiFrauD can detect at least 80% of suspicious traders and ideally can detect all suspicious traders in three datasets, much better than ANTICO, which is the best of the baselines. Also, in AUC, FiFrauD has surpassed the other baselines with a minimum difference of 5% and a maximum of 59%. In fact, most of the detected real-world traders have suspicious trading behavior during the given periods, which can be caught by FiFrauD easily and with high accuracy. From among the baselines, ANTICO has the highest AUC and recall; HoloScope has the lowest AUCs, and its recall is relatively lower than the others. As a result, compared with the other two methods, ANTICO, with a higher probability, can identify more fraudsters in the real-world environment. Therefore, FiFrauD outperforms the baselines on real-labeled data, as well.

6.3.2.2 Posteriori Analysis. To delve deeper into the behavior patterns of manipulators, we randomly select some detected traders from among the suspicious traders prosecuted by the TSE for

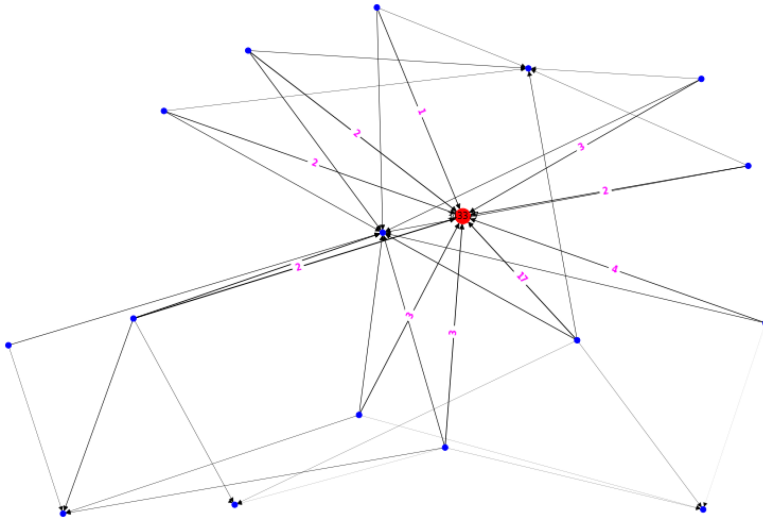


Fig. 5. Manipulation behavior pattern of the buyer detected by the TSE.

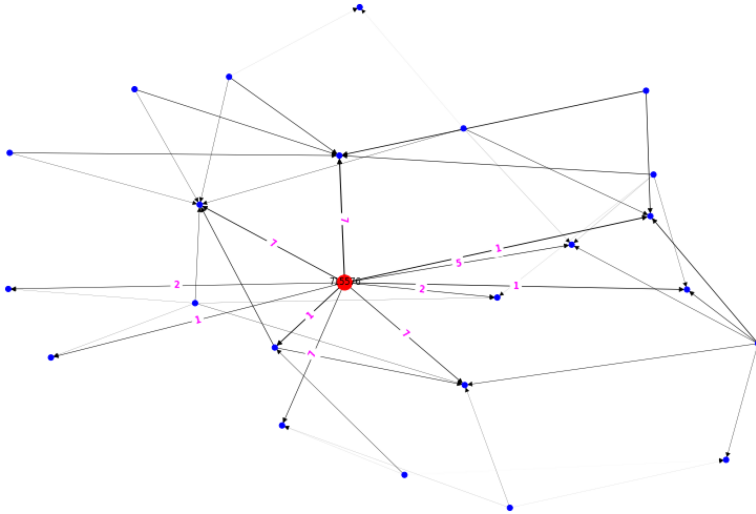


Fig. 6. Manipulation behavior pattern of the seller detected by the TSE.

the case study. Then, we extract their transactions from the data. Figures 5 and 6 show the behavior patterns of two selected traders, which are indicative of typical suspicious patterns (P1–P4). In these figures, the nodes are the traders and the red nodes show the selected traders, which are drawn larger in the figures for more clarity. The numbers written on the input or output edges show the edge frequency. The thickness of the edges is determined based on their suspiciousness rate.

A subgraph of a group suspected of fraud is drawn in Figure 5, in which one of the traders is detected as a fraudulent buyer by the TSE. According to the pattern drawn in the subgraph, a group colluded in buying and selling, and the confirmed fraudster in the group bought a very high volume of shares in 1 day (2,000,000 shares in 302 transactions). According to the transaction data, the detected fraudster, in 17 different transactions, has bought a volume of 400,000 shares

(per transaction) from the same person in the collusion group; that is, the bought a total number of 6,800,000 shares from one person in one day. This fraudster has started to submit buy orders with a very large volume suddenly in 15 minutes (sudden burst and drop), such that the individual submits 156 buy orders with high volumes in less than 15 minutes. This collusion group coherently begins to control their trades in terms of time, price, and volume. Thus, on one side, one person submits an order with a specified volume and price at a specific time, and on the other side, a buyer submits a buy order at the same time with the same volume and price, which indicates collusion between them. Another issue that is evident in the data is that the fraudulent person or persons would make some transactions with ordinary people and outside the detected collusion group to mislead the market supervisor; in other words, they try to camouflage. However, FiFrauD is resistant to camouflage and can detect such suspicious groups, which indicates that FiFrauD is appropriate for detecting fraudsters and colluders. In addition to the fraudster detected by the TSE, FiFrauD has the ability to detect other fraudsters, even the collusive groups and the suspicious transactions between them.

According to the transaction data, the suspicious person or persons have also bought a large number of shares from ordinary traders. Therefore, it can be said that the collusion group coherently submits buy and sell orders to encourage others to sell their shares in such a way that several colluding traders submit sell orders in a coherent manner and, on the other side, several submit buy orders at the same price and volume. The purpose of this behavior is to convince ordinary traders that the market is falling and to encourage the shareholders to sell their shares. The collusion group would buy whatever the ordinary traders are selling at their target price; that is, they try to take the shares out of ordinary investors' hands, which is considered fraudulent behavior.

The subgraph of another group suspected of fraud is drawn in Figure 6, in which one of the traders is detected as a fraudulent seller by the TSE. According to the data and the subgraph, the existing nodes and edges indicate fraud and collusion among the traders. For example, the trader's ID specified in the subgraph sold 12,240,000 shares in 274 transactions in one day and less than 30 minutes, and the majority of shares (11 million) were sold to 12 people, such that 4 million shares were sold to only one person in 20 transactions in 2 minutes. Data analysis shows coherence between buy and sell orders of the collusion group in terms of time, volume, and price, which is indicative of fraud and transactions between the group members. In addition, data analysis shows the occurrence of camouflage operation in the fraudulent group, so that 2,400,000 shares were sold to ordinary people in 148 transactions with the aim that the market supervisor would not notice the fraud or collusion. However, FiFrauD is resistant to camouflage and detects it.

The following four behaviors can be recognized based on the detected subgraphs. (1) *Divergency*: Most suspicious traders make their trades in several different transactions. (2) *Priority*: The traders at the forefront of the subgraphs (most of which are big shareholders), on average, make more transactions with suspicious traders. (3) *Heterophily*: Most of the detected traders trade with both suspicious shareholders and ordinary ones. FiFrauD can effectively exclude ordinary traders and transactions from the subgraphs. (4) *Hierarchy*: The more suspicious the subgraphs are, the more suspicious transactions by suspicious traders they will have. On the other hand, as the subgraph's suspiciousness decreases, the traders are inclined to spread the suspicious transactions and do more normal ones.

6.3.3 Q3. Scalability. We use the STK3_6 Month dataset (see Table 5) to analyze the runtime. Figure 7 illustrates that FiFrauD's runtime scales linearly with the number of stream edges, aligning with the time complexity described in Section A.2 of the appendix. As a result, FiFrauD can effectively detect anomalies in a graph stream, even in the largest graphs.

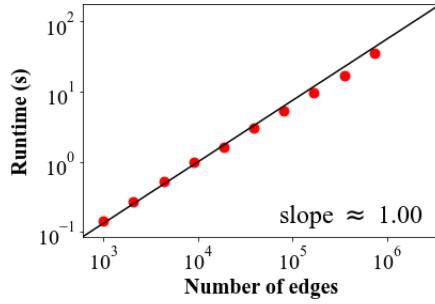


Fig. 7. FiFrauD runs in time linear to the number of edges in a transaction stream.

7 DISCUSSION

Market manipulation through buying and selling, whose intensity is increasing day by day, has become a significant challenge in financial e-markets. As protecting regular investors from such manipulations is crucial, researchers have tried to develop effective detection methods to quickly identify suspicious trading behaviors among vast trades, leading to good results. But it seems that the studies conducted do not fully and deeply investigate the problem of manipulation detection, such that this field still faces serious challenges, including the ineffectiveness of existing methods for the complex and streaming environment of e-markets. Therefore, despite the efforts made, it is still difficult to quickly and accurately identify market manipulations. It is still an open problem, and there is a need for broader and deeper research in this field. In this regard, to solve this problem and alleviate the existing challenges, the present research proposes a novel detection framework, called FiFrauD, which provides significant findings. In the following, we discuss the most important of these findings.

Experimental results on real data with injected labels indicate that FiFrauD can detect different patterns of manipulative behaviors with high accuracy. In fact, due to the comprehensive use of different signals in its detection framework, FiFrauD consistently outperforms all baseline methods in detecting different manipulative behavior patterns. In the highest fraudulent density, it led to improvements of 12% to 39% in F-measure, 4% to 46% in AUC, 1% to 42% in precision, 13% to 49% in recall, and 1% to 7% in accuracy. More importantly, in the lowest fraudulent density, i.e., in situations in which spotting fraudsters is challenging, FiFrauD still performs well, leading to improvements of 39% to 63% in F-measure, 9% to 47% in AUC, 6% to 45% in precision, 46% to 71% in recall, and 1% to 4% in accuracy. Even in situations in which the number of fraudsters is small and different methods fail to detect them, based on these results, FiFrauD is able to spot fraudsters with different behavior patterns with high detection accuracy and low false alarms. Also, when fraudsters try to avoid detection by making camouflage transactions, FiFrauD can still catch them. Therefore, FiFrauD has led to a significant improvement in the detection accuracy of fraudsters and a reduction in false alarms.

Experimental results to spot real-world fraudulent traders identified by market supervisors confirm the proper performance of FiFrauD in all five datasets. By achieving improvements of 5% to 59% in AUC and 20% to 67% in recall, it has been able to surpass all baseline methods. Therefore, FiFrauD has also outperformed other detection methods in spotting real-world fraudulent traders and has been able to spot them with high accuracy. In addition, FiFrauD is scalable, can catch fraudulent traders in linear runtime, and can provide interpretable results. According to the results of the theoretical analysis, FiFrauD provides provable theoretical bounds for the duration of the manipulation as well as the number of transactions. Therefore, it can obstruct fraudsters

and increase their time cost and effort needed to manipulate a market. Based on this, it can be concluded that the proposed approach of this research is suitable for the complex and streaming environment of financial e-markets and can satisfy the needs of market supervisors.

In summary, this study is significantly different from the existing studies and deals with novel concepts, the most important of which are as follows. First, instead of processing static data, it deals with online detection in a transaction stream. Second, it does not detect in a black box way. Rather, by analyzing manipulative behaviors and their key characteristics, it provides novel suspiciousness metrics. Third, it pays attention to the relational nature of trading data and considers existing relationships between traders. Fourth, in order to confirm the performance of the proposed method in detecting various abnormal behaviors, experiments are designed and implemented based on both real and synthetic labels. These properties in the detection model lead to better performance of the proposed approach in terms of both detection accuracy and time complexity. In general, this research contributes to the literature enrichment and knowledge expansion on market manipulation and financial fraud, as well as filling the research gap emphasized by review studies. By satisfying the needs of the market legislators and supervisors, it helps them to understand and recognize the manipulation methods as well as how to detect them, and enables the successful detection of suspicious transactions and traders. This makes financial markets a reliable and fair environment where prices reflect real supply and demand, thus leading to the satisfaction of market participants and supervisors and the improvement of the economy at the national and international levels.

8 CONCLUSION

This study aims to detect manipulation in e-markets. Based on an unsupervised graph-based approach, it considers manipulation detection as a problem of discovering dense subgraphs. In this regard, it detects suspicious traders and trading patterns by optimizing density signals in a stream of attributed dynamic graphs. Specifically, it reveals the traits of fraudulent traders in the market and proposes a novel metric from this perspective, namely, the intensiveness of trades, high involvement, sudden bursts and drops, and high cohesiveness. It develops FiFrauD for searching suspicious subgraphs by greedily optimizing the proposed metric. The results of experiments on real and semi-real datasets confirm that FiFrauD can spot fraudulent traders in linear runtime and provide interpretable results. This study is a pioneering work that detects market manipulations by identifying anomalies in dynamic graphs and employing dense subgraph mining. The main task for the future is to improve the search process and introduce more properties, including the duration between consecutive transactions by a trader, to enhance performance. Also, detecting order-based manipulations in e-markets and the implementation of the proposed method on the data of other e-markets, such as the cryptocurrency market, can be defined as future work for detecting financial frauds.

REFERENCES

- [1] V. Dalko and M. H. Wang. 2020. High-frequency trading: Order-based innovation or manipulation? *J Bank Regu.* 21, 4 (2020 Dec 1), 289–98.
- [2] D. Diaz, B. Theodoulidis, and P. Sampaio. 2011. Analysis of stock market manipulations using knowledge discovery techniques applied to intraday trade prices. *Expert Systems with Applications* 38, 10 (2011 Sep 15), 12757–12771.
- [3] C. Alexander and D. Cumming. 2020. *Corruption and Fraud in Financial Markets: Malpractice, Misconduct and Manipulation*. John Wiley & Sons (2020), 624.
- [4] K. Golmohammadi and O. R. Zaiane. 2015. Time series contextual anomaly detection for detecting market manipulation in stock market. In: *2015 IEEE International Conference on Data Science and Advanced Analytics (DSAA)* (2015), 1–10.
- [5] T. J. Putninš. 2018. An overview of market manipulation [Internet]. Rochester, NY: Social Science Research Network; 2018 Oct [cited 2019 Dec 6]. Report No.: ID 3398258. Retrieved from <https://papers.ssrn.com/abstract=3398258>

- [6] M. Siering, B. Clapham, O. Engel, and P. Gomber. 2017. A taxonomy of financial market manipulations: establishing trust and market integrity in the financialized economy through automated fraud detection. *J Inf Technol* 32, 3 (2017 Sep 1), 251–269.
- [7] S. Khodabandehlou and S. Alireza Hashemi Golpayegani. 2022. Market manipulation detection: A systematic literature review. *Expert Systems with Applications* 210 (2022 Dec 30), 118330.
- [8] Y. Cao, Y. Li, S. Coleman, A. Belatreche, and T. M. McGinnity. 2016. Detecting wash trade in financial market using digraphs and dynamic programming. *IEEE Transactions on Neural Networks and Learning Systems* 27, 11 (2016 Nov), 2351–2363.
- [9] Q. Wang, W. Xu, X. Huang, and K. Yang. 2019. Enhancing intraday stock price manipulation detection by leveraging recurrent neural networks with ensemble learning. *Neurocomputing* 28 (2019 Jun), 347, 46–58.
- [10] J. Zhai, Y. Cao, Y. Yao, X. Ding, and Y. Li. 2017. Coarse and fine identification of collusive clique in financial market. *Expert Systems with Applications* 69 (2017 Mar 1), 225–238.
- [11] J. Zhai, Y. Cao, Y. Yao, X. Ding, and Y. Li. 2017. Computational intelligent hybrid model for detecting disruptive trading activity. *Decision Support Systems* 93 (2017 Jan 1), 26–41.
- [12] P. Chullamonthon and P. Tangamchit. 2023. Ensemble of supervised and unsupervised deep neural networks for stock price manipulation detection. *Expert Systems with Applications*. 220 (2023 Jun 15), 119698.
- [13] T. Leangarun, P. Tangamchit, and S. Thajchayapong. 2021. Stock price manipulation detection using deep unsupervised learning: The case of Thailand. *IEEE Access* 9 (2021), 106824–106838.
- [14] X. Q. Sun, H. W. Shen, X. Q. Cheng, and Y. Zhang. 2017. Detecting anomalous traders using multi-slice network analysis. *Physica A: Statistical Mechanics and its Applications* 1 (2017 May), 473, 1–9.
- [15] N. C. Uslu and F. Akal. 2022. A machine learning approach to detection of trade-based manipulations in Borsa Istanbul. *Comput Econ* 60, 1 (2022 Jun 1), 25–45.
- [16] S. Dhawan, S. C. R. Gangireddy, S. Kumar, T. Chakraborty. 2019. Spotting collective behaviour of online frauds in customer reviews. In *Proceedings of the 28th International Joint Conference on Artificial Intelligence (IJCAI'19)*, AAAI Press, Macao, 245–251.
- [17] S. Liu, B. Hooi, and C. Faloutsos. 2019. A contrast metric for fraud detection in rich graphs. *IEEE Transactions on Knowledge and Data Engineering* 31, 12 (2019 Dec), 2235–2248.
- [18] S. Pillai and D. R. A. Sharma. 2023. Hybrid unsupervised web-attack detection and classification – A deep learning approach. *Computer Standards & Interfaces* 86 (2023 Aug 1), 103738.
- [19] S. Azadifar, M. Rostami, K. Berahmand, P. Moradi, and M. Oussalah. 2022. Graph-based relevancy-redundancy gene selection method for cancer diagnosis. *Computers in Biology and Medicine* 147, (2022 Aug 1), 105766.
- [20] S. Chen, W. Susilo, Y. Zhang, B. Yang, and M. Zhang. 2023. Privacy-preserving anomaly counting for time-series data in edge-assisted crowdsensing. *Computer Standards & Interfaces* 85 (2023 Apr 1), 103707.
- [21] S. Huang, X. Yang, L. He, X. Hao, and W. Ren. 2023. An analysis model for detecting misbehaviors in anonymous cryptocurrency. *Computer Standards & Interfaces* 83 (2023 Jan 1), 103669.
- [22] Z. H. Janjua, M. Vecchio, M. Antonini, and F. Antonelli. 2019. IRESE: An intelligent rare-event detection system using unsupervised learning on the IoT edge. *Engineering Applications of Artificial Intelligence* 84 (2019 Sep 1), 41–50.
- [23] M. Ahmed, A. Naser Mahmood, and J. Hu. 2019. A survey of network anomaly detection techniques. *Journal of Network and Computer Applications* 60 (2016 Jan 1), 19–31.
- [24] E. W. T. Ngai, Y. Hu, Y. H. Wong, Y. Chen, and X. Sun. 2011. The application of data mining techniques in financial fraud detection: A classification framework and an academic review of literature. *Decision Support Systems* 50, 3 (2011 Feb 1), 559–569.
- [25] I. Sadgali, N. Sael, and F. Benabbou. 2019. Performance of machine learning techniques in the detection of financial frauds. *Procedia Computer Science* 148 (2019 Jan 1), 45–54.
- [26] J. O. Sinayobye, F. Kiwanuka, and S. Kaawaase Kyanda. 2018. A state-of-the-art review of machine learning techniques for fraud detection research. In *2018 IEEE/ACM Symposium on Software Engineering in Africa (SEiA)* (2018). 11–19.
- [27] J. West and M. Bhattacharya. 2016. Intelligent financial fraud detection: A comprehensive review. *Computers & Security* 57 (2016 Mar 1), 47–66.
- [28] M. Jiang, A. Beutel, P. Cui, B. Hooi, and S. Yang. 2015. Faloutsos C. A general suspiciousness metric for dense blocks in multimodal data. In *2015 IEEE International Conference on Data Mining*. 781–6.
- [29] T. Chen and C. Tsourakakis. 2022. AntiBenford subgraphs: Unsupervised anomaly detection in financial networks. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining [Internet]*. New York, NY, USA: Association for Computing Machinery; 2022 [cited 2023 Apr 17]. 2762–2770. (KDD'22). Retrieved from <https://doi.org/10.1145/3534678.3539100>
- [30] X. Li, S. Liu, Z. Li, X. Han, C. Shi, B. Hooi et al. 2020. FlowScope: Spotting money laundering based on graphs. *Proceedings of the AAAI Conference on Artificial Intelligence* 34, 04 (2020 Apr 3), 4731–4738.

- [31] Y. Ji, Z. Zhang, X. Tang, J. Shen, X. Zhang, and G. Yang. 2022. Detecting cash-out users via dense subgraphs. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining [Internet]*. New York, NY, USA: Association for Computing Machinery; 2022 [cited 2023 Apr 17]. 687–697. (KDD'22). Retrieved from <https://doi.org/10.1145/3534678.3539252>
- [32] S. Liu, B. Hooi, and C. Faloutsos. 2017. HoloScope: Topology-and-spike aware fraud detection. In *Proceedings of the 2017 ACM on Conference on Information and Knowledge Management [Internet]*. New York, NY, USA: Association for Computing Machinery; 2017 [cited 2023 Apr 17]. 1539–1548. (CIKM'17). Retrieved from <https://dl.acm.org/doi/10.1145/3132847.3133018>
- [33] S. Khodabandehlou and M. Zivari Rahman. 2017. Comparison of supervised machine learning techniques for customer churn prediction based on analysis of customer behavior. *Journal of Systems and Information Technology* 19, 1/2 (2017 Jan 1), 65–93.
- [34] K. Golmohammadi, O. R. Zaiane, and D. Díaz. 2014. Detecting stock market manipulation using supervised learning algorithms. In *2014 International Conference on Data Science and Advanced Analytics (DSAA)*. (2014). 435–441.
- [35] R. Sheikhpour, K. Berahmand, and S. Forouzandeh. 2023. Hessian-based semi-supervised feature selection using generalized uncorrelated constraint. *Knowledge-Based Systems* 269 (2023 Jun 7), 110521.
- [36] T. Leangarn, P. Tangamchit, and S. Thajchayapong. 2018. Stock price manipulation detection using generative adversarial networks. In *2018 IEEE Symposium Series on Computational Intelligence (SSCI)*. (2018). 2104–2111.
- [37] B. Rizvi, A. Belatreche, and A. Bouridane. 2019. A dendritic cell immune system inspired approach for stock market manipulation detection. In *2019 IEEE Congress on Evolutionary Computation (CEC) [Internet]*. Wellington, New Zealand: IEEE; 2019 [cited 2020 Apr 23]. 3325–3332. Retrieved from <https://ieeexplore.ieee.org/document/8789938/>
- [38] F. B. Shi, X. Q. Sun, and H. W. Shen. 2019. Cheng XQ. Detect colluded stock manipulation via clique in trading network. *Physica A: Statistical Mechanics and its Applications* 513 (2019 Jan 1), 565–571.
- [39] G. K. Palshikar and M. M. Apte. 2008. Collusion set detection using graph clustering. *Data Min Knowl Disc* 16, 2 (2008 Apr 1), 135–164.
- [40] E. Martínez Miranda, S. Phelps, and M. J. Howard. 2019. Order flow dynamics for prediction of order cancelation and applications to detect market manipulation. *High Frequency* 2, 1 (2019 Feb 1), 4–36.
- [41] W. Chen, Y. Xu, Z. Zheng, Y. Zhou, J. E. Yang, and J. Bian. 2019. Detecting “pump dump schemes” on cryptocurrency market using an improved apriori algorithm. In *2019 IEEE International Conference on Service-Oriented System Engineering (SOSE)* (2019). 293–2935.
- [42] L. Akoglu, M. McGlohon, and C. Faloutsos. 2010. Oddball: Spotting anomalies in weighted graphs. In: M. J. Zaki, J. X. Yu, B. Ravindran, V. Pudi, editors. *Advances in Knowledge Discovery and Data Mining (Lecture Notes in Computer Science)*. Berlin, Springer; 2010. 410–421.
- [43] F. Braun, O. Caelen, E. N. Smirnov, S. Kelk, and B. Lebichot. 2017. Improving card fraud detection through suspicious pattern discovery. In: Benferhat S, Tabia K, Ali M, editors. *Advances in Artificial Intelligence: From Theory to Practice. (Lecture Notes in Computer Science)*. Cham: Springer International Publishing; 2017. 181–190.
- [44] V. Herrera-Semenets, N. Acosta-Mendoza, and A. Gago-Alonso. 2015. A framework for intrusion detection based on frequent subgraph mining. In *The 2nd SDM Workshop on Mining Networks and Graphs: A Big Data Analytic Challenge (SDM-Networks 2015)*, Vancouver, BC.
- [45] H. Nilforoshan and N. Shah. 2019. SliceNDice: Mining suspicious multi-attribute entity groups with multi-view graphs. In *2019 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*. 351–363.
- [46] S. Zhang and J. Li. 2023. KNN classification with one-step computation. *IEEE Transactions on Knowledge and Data Engineering* 35, 3 (2023 Mar), 2711–2723.
- [47] S. Zhang, J. Li, and Y. Li. 2023. Reachable distance function for KNN classification. *IEEE Trans on Knowl and Data Eng* 35, 7 (2023 Jul 1), 7382–7396.
- [48] S. Zhang, J. Li, W. Zhang, and Y. Qin. 2022. Hyper-class representation of data. *Neurocomputing* 503 (2022 Sep 7), 200–218.
- [49] G. Zhang, D. Cheng, G. Yuan, and S. Zhang. 2024. Learning fair representations via rebalancing graph structure. *Information Processing & Management* 61, 1 (2024 Jan 1), 103570.
- [50] G. Zhang, D. Cheng, and S. Zhang. 2023. FPGNN: Fair path graph neural network for mitigating discrimination. *World Wide Web* 26, 5 (2023 Sep 1), 3119–3136.
- [51] H. Liu, X. Li, and J. Li, and S. Zhang. 2018. Efficient outlier detection for high-dimensional data. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 48, 12 (2018 Dec), 2451–2461.
- [52] S. Zhang. 2022. Challenges in KNN classification. *IEEE Transactions on Knowledge and Data Engineering* 34, 10 (2022 Oct), 4663–4675.
- [53] S. Zhang, X. Li, M. Zong, X. Zhu, and R. Wang. 2018. Efficient kNN classification with different numbers of nearest neighbors. *IEEE Transactions on Neural Networks and Learning Systems* 29, 5 (2018 May), 1774–1785.
- [54] K. Shin, T. Eliassi-Rad, and C. Faloutsos. 2016. CoreScope: Graph mining using k-core analysis — patterns, anomalies and algorithms. In *2016 IEEE 16th International Conference on Data Mining (ICDM)*. 469–478.

- [55] A. Beutel, W. Xu, V. Guruswami, C. Palow, and C. Faloutsos. 2013. CopyCatch: stopping group attacks by spotting lockstep behavior in social networks. In *Proceedings of the 22nd International Conference on World Wide Web [Internet]*. New York, NY, USA: Association for Computing Machinery; 2013 [cited 2023 Apr 17]. 119–130. (WWW'13). Retrieved from <https://doi.org/10.1145/2488388.2488400>
- [56] B. Hooi, H. A. Song, A. Beutel, N. Shah, K. Shin, and C. Faloutsos. 2016. FRAUDAR: Bounding graph fraud in the face of camouflage. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining [Internet]*. New York, NY, USA: Association for Computing Machinery; 2016 [cited 2023 Apr 17]. 895–904. (KDD'16). Retrieved from <https://dl.acm.org/doi/10.1145/2939672.2939747>
- [57] S. Bhatia, M. Wadhwa, K. Kawaguchi, N. Shah, and P. S. Yu. 2023. Hooi B. Sketch-based anomaly detection in streaming graphs. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining [Internet]*. New York, NY, USA: Association for Computing Machinery; 2023 [cited 2023 Nov 8]. 93–104. (KDD'23). Retrieved from <https://dl.acm.org/doi/10.1145/3580305.3599504>
- [58] B. A. Prakash, M. Seshadri, A. Sridharan, S. Machiraju, and C. Faloutsos. 2009. EigenSpokes: surprising patterns and scalable community chipping in large graphs. In *2009 IEEE International Conference on Data Mining Workshops*. 290–295.
- [59] Association of Certified Fraud Examiners. 2002. Report to the Nation [Internet]. [cited 2020 Jul 2]. Retrieved from <https://www.acfe.com/article.aspx?id=4294968212>
- [60] Oxford Concise English Dictionary. 2011. *Concise Oxford English Dictionary: Main Edition*. 12th ed., Oxford University Press, New York, NY, 1728 pages.
- [61] Financial Crimes Report 2007 — FBI [Internet]. [cited 2020 Jul 1]. Retrieved from https://www.fbi.gov/stats-services/publications/fcs_report2007
- [62] European Commission. Investment services and regulated markets — Markets in financial instruments directive (MiFID). European Commission. [cited 2020 Jul 2]. Retrieved from https://ec.europa.eu/info/business-economy-euro/banking-and-finance/financial-markets/securities-markets/investment-services-and-regulated-markets-markets-financial-instruments-directive-mifid_en
- [63] Commodity Futures Trading Commission | CFTC [Internet]. [cited 2020 Jul 2]. Retrieved from <https://www.cftc.gov/>
- [64] F. Allen and D. Gale. 1992. Stock-price manipulation. *Rev Financ Stud* 5, 3 (1992 Jul 1), 503–529.
- [65] H. A. Sturges. 1926. The choice of a class interval. *Journal of the American Statistical Association* 21, 153 (1926 Mar 1), 65–66.
- [66] D. Freedman and P. Diaconis. 1981. On the histogram as a density estimator: L2 theory. *Z Wahrscheinlichkeitstheorie verw Gebiete* 57, 4 (1981 Dec 1), 453–476.
- [67] M. Charikar. 2000. Greedy approximation algorithms for finding dense components in a graph. In: Jansen K, Khuller S, editors. *Approximation Algorithms for Combinatorial Optimization. (Lecture Notes in Computer Science)*. Berlin, Springer. 84–95.

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